

BREAKHOOK · REQUIREMENTS

BreakHook requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/breakhook		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-03	Amish Chadha	R1 stand-off wording restated after Amish's round-2 decision 5A (BHK-DDR-004): prop holder at 2.9 m during placement only; confirmed in the TRL 4 reach trial
0.3	2026-10-03	Amish Chadha	Fork prop at mid-length added under Amish's decision 6A (BHK-DDR-003); status of R1, R5, R8, R10 and R14 updated from BHK-CAL-001 v0.2; targets unchanged
0.2	2026-10-03	Amish Chadha	TRL 2 and TRL 3; targets confirmed under Amish's 2026-10-03 pre-approval (BHK-DDR-001); R11 to R14 added; status from BHK-CAL-001
0.1	2026-09-30	Amish Chadha	Initial scaffold

BreakHook requirements

Fourteen requirements, each with a measurable target. At TRL 3 they are checked by calculation (BHK-CAL-001); verification by test is TRL 4 work. Ten are met on paper; R1's hook placement is met with the fork prop, and the prop holder's 2.9 m placement-only stand-off is accepted by Amish (the TRL 4 reach trial confirms it); R6 and R13 are met with no margin, and R7 and R9 can only be shown by trial.

On 2026-10-03 Amish chose option A on every requirement decision put to him ("1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"). For BreakHook, decision 6A adds a light fork prop that supports the pole at mid-length, carried separately so the pole and hook head stay under 8 kg (BHK-DDR-003). No target changes.

Safety: BreakHook pulls a structure down with a rope. Every requirement assumes the safety rules of BHK-PRC-001: the dwelling is checked empty, the fall zone is clear and taped, nobody is within 3 m of an overhead line, and the pole is withdrawn before the haul. A requirement is never met by relaxing those rules.

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R1	Reach of the hook	Places the hook at a height of 3 m (10 ft) with the pole hands (front hand) 4 m (13 ft) from the wall; the prop holder may stand 2.9 m from the wall during placement only (about 25 s) and withdraws with the pole before the haul	Reach trial on the test frame, with the prop holder's distance from the wall and the time at 2.9 m measured	Hook placement met on paper with the fork prop at mid-length: droop about 0.27 m (0.66 m by hand), rear hand 5.06 m from the wall (CAL [F13] to [F20]); the prop holder stands about 2.9 m from the wall for the placement only (about 25 s), accepted by Amish (5A, round 2), and withdraws with the pole; the TRL 4 reach trial confirms it
R2	Hook and rope working pull	3 kN (675 lbf) working, every hook head proof-loaded to 6 kN (1,350 lbf) before issue	Proof load with CalRig or at a rigging shop	Met: least factor 2.3 on yield at proof (CAL [C3])
R3	Hauling distance	Nearest hauler at least 1.5 times the structure height from the wall, and never closer than 10 m	Field layout check	Met: 10.8 m (CAL [B1])
R4	Pole electrical insulation	The handled length is foam-filled fibreglass carrying the maker's dielectric test certificate to ASTM F711 for live-line tool tube	Certificate check on receipt; no field electrical test	Met by specification; confirm the certificate when bought
R5	Pole and hook mass	8 kg (18 lb) or less	Weigh	Met: 7.82 kg (CAL [F10]); the 1.08 kg fork prop is carried separately (CAL [F22])
R6	Deployment time	From store to hook set in 3 min or less with four people, store within 100 m	Timed drill	Met on estimate with no margin: 3.0 min (CAL [I1])
R7	Time to bring down a test dwelling frame	Under 2 min from hook set, single-storey timber and sheet frame	Timed trial on a purpose-built test frame, never an occupied area	Estimate only: about 1.2 min (CAL [I2])
R8	Section length for narrow paths	No piece longer than 2 m (6.5 ft)	Measure	Met: 1.95 m (CAL [F12]); fork prop 1.24 m closed
R9	Storage life	No loss of function after 12 months in the rack	Inspection and repeat proof load after exposure	Not verifiable at TRL 3; material choices only
R10	Cost	Community kit (two sets, rack, cards, tape, gloves) against the USD 2,000 value-engineering target	Costed bill of materials	USD 1,170 with two fork props, USD 830 under the target
R11	Beam size the hook takes	Timber up to 100 mm deep and 120 mm wide	Fit check on timber offcuts	Met: 117 mm deep, 128 to 150 mm wide (CAL [C8] to [C10])
R12	Pole release			Met by design; set by trial (CAL [G1])

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
		The pole comes free of the hook head with a pull of about 50 N, so it is withdrawn before the haul	Push-off check with a spring balance	
R13	Effort per hauler	300 N or less per person at the working pull	Count of toggles and crew	Met at the limit: ten haulers at 300 N (CAL [A5])
R14	Insulated length	At least 3.0 m with no metal between the hook head and the hand band	Measure	Met: 3.67 m (CAL [H1]); 3.27 m from the socket mouth to the prop holder's hand band (CAL [H3])

■ Requirements at risk

- **R1:** the fibreglass tube is flexible. The fork prop at mid-length cuts the droop from about 0.66 m to 0.27 m and takes the lift off the pole team, but its holder stands about 2.9 m from the wall while the hook is set, for about 25 s. Amish accepted that for placement only (round 2, 5A: "i agree with all the 46 recommendations you provided. please proceed.") and R1 is restated accordingly: the 4 m applies to the pole hands, and the holder withdraws with the pole before the haul. The reach trial at TRL 4 confirms the distance and the time; the measured droop is an item to confirm.
- **R6 and R13:** both sit exactly on their targets, so a slower carry or a weaker crew misses them. The community fire plan sets the store within 100 m and calls for ten haulers.
- **R5:** 0.18 kg of margin; heavier paint or a thicker tube wall would use it up.

■ Assumptions

- Typical dwellings are single storey with timber or light steel frames that a crew can pull down with a hook (BHK-CAL-001, assumption 1).
- Communities already accept firebreaks as a response and can agree rules on use; the kit's governance (sealed rack, named keyholders, use log) supports those rules but does not replace them.
- The fire service supports community action in the first minutes.
- Test frames are built to represent real dwellings; occupied homes are never used for tests.